Explore Packets and Frames | Study Guide

Mission: capture a simulated HTTP POST request, apply a display filter, and inspect the Ethernet II, IPv4, TCP, and HTTP protocol layers.

REFERENCE IMAGE A | Data Encapsulation

L7	HTTP Application Data
L4	TCP Segment | Src 51514 / Dst 80
L3	IPv4 Packet | 192.168.10.25 to 203.0.113.80
L2	Ethernet Frame | MAC addresses + type 0x0800
L1	Bits on the network medium

REFERENCE IMAGE B | Wireshark-style Protocol Tree

Frame 4: 483 bytes on wire

Ethernet II: Src 00:16:3e:12:34:56 | Dst 00:aa:bb:cc:dd:01

Internet Protocol Version 4: Header 20 bytes | Protocol TCP

Transmission Control Protocol: 51514 -> 80 | Flags PSH, ACK

Hypertext Transfer Protocol: POST /training HTTP/1.1

Learning objectives

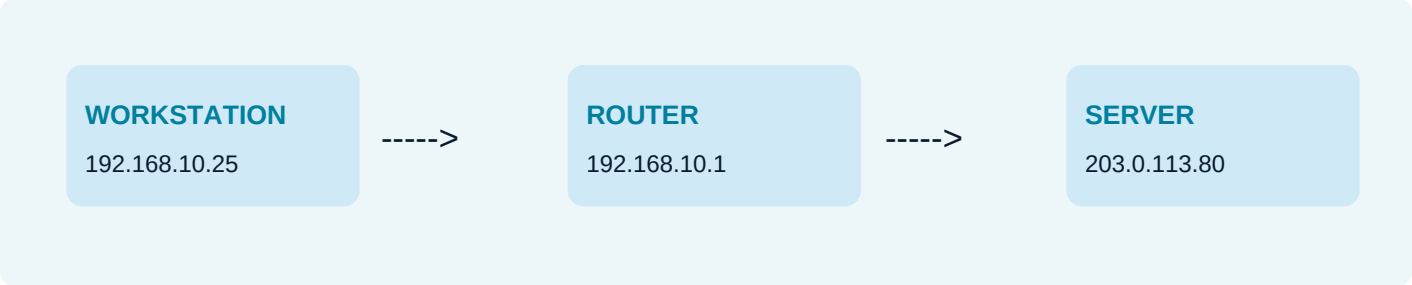
- Recognize the difference between Ethernet frames, IP packets and TCP segments.
- Identify MAC addresses, IP addresses, TCP ports and TCP flags.
- Explain why routing normally replaces the Layer 2 header but preserves endpoints.

Vocabulary | Know What You See

Ethernet frame	Layer 2 data unit with source/destination MAC addresses.
IP packet	Layer 3 unit with source/destination IP addresses.
TCP segment	Layer 4 data unit with ports, sequencing and flags.
Encapsulation	Adding protocol headers as data moves down a network stack.
Decapsulation	Removing headers as data moves up the receiving stack.
MAC address	Link-layer hardware identifier for the local network hop.
IP address	Logical Layer 3 endpoint identifier used for routing.
Display filter	Expression used to show only matching packets in a capture.
HTTP POST	Method that submits request content to a web server.
IPv4 IHL	Internet Header Length, measured in four-byte units.
TTL	IPv4 time-to-live; routers decrease this value.
TCP port	16-bit identifier for an application endpoint.
TCP PSH/ACK	Push and acknowledgment flags in TCP.
EtherType 0x0800	Ethernet type field meaning an IPv4 packet is encapsulated.
Wireshark	Widely used graphical network protocol analyzer.
FCS	Frame Check Sequence for error detection; often not recorded in captures.

Visual Reference | Addressing & Troubleshooting

REFERENCE IMAGE C | Frame at the first routed hop



Before routing: Ethernet destination is the gateway MAC, not the remote server MAC. A router removes the original Ethernet header and constructs a new link-layer header for the outgoing interface. Without NAT, the IPv4 endpoint addresses normally remain unchanged.

Example packet fields to inspect

Ethernet source	00:16:3e:12:34:56
Ethernet destination	00:aa:bb:cc:dd:01 (gateway)
IPv4 source / destination	192.168.10.25 / 203.0.113.80
IPv4 header length	IHL 5 x 4 = 20 bytes
IPv4 Protocol	6 = TCP
TCP destination port	80 = HTTP
TCP flags for POST	PSH, ACK

Troubleshooting note

A display filter such as http.request.method == POST isolates matching decoded HTTP packets. If none appear, verify that traffic was generated and that the selected traffic is actually unencrypted HTTP. HTTPS payload content is not normally visible as plain HTTP without decryption.

Practice | Procedure & Review

Guided practice (complete in the HTML lab)

1. Start capture on Ethernet0 and generate an HTTP POST.
2. Stop capture and apply: http.request.method == POST
3. Select the POST packet and expand Ethernet II.
4. Inspect IPv4 version, header length, protocol and endpoints.
5. Inspect TCP source/destination ports and flags.
6. Verify encapsulation and routed-hop behavior.
7. Answer five review questions and print your verified report.

Review questions

1. What is the difference between a frame and a packet?
2. Why is the local destination MAC not the web server MAC?
3. Which IPv4 field identifies TCP?
4. What does a destination port of 80 mean?
5. How many bytes is an IPv4 header when IHL = 5?

Answer check

1. Frame is Layer 2; packet is Layer 3.
2. Off-subnet traffic goes to the gateway's MAC.
3. Protocol field value 6.
4. HTTP over TCP (unencrypted in this example).
5. 20 bytes.

Student: _____ Date: _____

MockTIA | NETWORK 4.1.6

OSI 7-Layer Visual Reference

Click the matching colored layers in the interactive HTML lab for deeper examples.

7	**Application** HTTP request and DNS	**Data**
6	**Presentation** Encoding / encryption concepts	**Data**
5	**Session** Session management concepts	**Data**
4	**Transport** TCP / UDP; ports, flags	**Segment / datagram**
3	**Network** IPv4 addresses, routing, TTL	**Packet**
2	**Data Link** Ethernet MAC addresses; switches	**Frame**
1	**Physical** Electrical / optical / radio signals	**Bits**

TCP/IP mapping:

Application = OSI L5-L7 | Transport = L4 | Internet = L3 | Link = L1-L2

Remember: OSI layers 5 and 6 are conceptual, not separate Ethernet packet headers.

MockTIA | NETWORK 4.1.6

Encapsulation: Data to Bits

Follow the same HTTP POST example as the packet-capture simulator.

L7 HTTP application data

HTTP POST

v

L4 TCP segment

TCP header | HTTP data

v

L3 IPv4 packet

IPv4 header | TCP header | HTTP data

v

L2 Ethernet frame

Ethernet | IPv4 | TCP | DATA | FCS

v

L1 Bits over media

01001011 10110010 00110101 ...

Routed hop (no NAT):

Ethernet MAC header changes | IP endpoints stay | IPv4 TTL decreases